NEWTON SCHOOL

PRE-CALCULUS | ANSWER KEY

Solving Trigonometric Equations

Student: Anne Takemoto **Date:** 2026-05-12

Note: *All solutions in* $[0, 2\pi)$ *unless stated otherwise. General form given where applicable.*

Mr. Kenny's Key Point

Two big traps in trig equations:

1. **Always check the range** ($|\sin| \leq 1$, $|\cos| \leq 1$) — extraneous roots like $\cos x = 3$ must be rejected.

2. For **multiple-angle** equations ($2x$, $x/2$, etc.), find ALL solutions in the wider domain first, then translate back to x.

Part A — Direct Equations & Factoring

1. $2\sin x - 1 = 0$ **on** $[0, 2\pi)$

Isolate: $2\sin x = 1 \;\Rightarrow\; \sin x = \dfrac{1}{2}$

Reference angle: $\dfrac{\pi}{6}$

Sine is positive in QI and QII: $x = \dfrac{\pi}{6}, \dfrac{5\pi}{6}$

$$\boxed{x = \frac{\pi}{6}, \frac{5\pi}{6}}$$

2. $\tan x + 1 = 0$ **on** $[0, 2\pi)$

> Isolate: $\tan x = -1$
>
> Reference angle: $\dfrac{\pi}{4}$
>
> Tangent is negative in QII and QIV:
> $x = \pi - \dfrac{\pi}{4} = \dfrac{3\pi}{4}, \quad x = 2\pi - \dfrac{\pi}{4} = \dfrac{7\pi}{4}$

$$\boxed{x = \frac{3\pi}{4}, \frac{7\pi}{4}}$$

3. $2\sin\theta\cos\theta + \cos\theta = 0$ **on** $[0, 2\pi)$

> Factor: $\cos\theta\,(2\sin\theta + 1) = 0$
>
> **Case 1:** $\cos\theta = 0 \;\Rightarrow\; \theta = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$
>
> **Case 2:** $2\sin\theta + 1 = 0 \;\Rightarrow\; \sin\theta = -\dfrac{1}{2}$
>
> Sine is negative in QIII and QIV:
> $\theta = \pi + \dfrac{\pi}{6} = \dfrac{7\pi}{6}, \quad \theta = 2\pi - \dfrac{\pi}{6} = \dfrac{11\pi}{6}$

$$\boxed{\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}}$$

4. $\sin^2 x - \cos^2 x = 0$ **on** $[0, 2\pi)$

> Rewrite: $\sin^2 x = \cos^2 x$
>
> Divide by $\cos^2 x$ (valid wherever $\cos x \neq 0$, checked separately):
> $\tan^2 x = 1 \;\Rightarrow\; \tan x = \pm 1$
>
> Reference angle: $\dfrac{\pi}{4}$. Tangent is ± 1 in all four quadrants.
>
> $x = \dfrac{\pi}{4}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}, \dfrac{7\pi}{4}$
>
> *Check* $\cos x = 0$*:* $x = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$ *give* $\sin^2 x - \cos^2 x = 1 \neq 0$. *Not solutions.*

$$\boxed{x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}}$$

Part B — Quadratic in a Trig Function

5. $2\cos^2 x - 7\cos x + 3 = 0$ **on** $[0, 2\pi)$

Let $u = \cos x$: $2u^2 - 7u + 3 = 0$

Factor: $(2u-1)(u-3) = 0$

$u = \dfrac{1}{2}$ OR $u = 3$ **REJECT:** $|\cos x| \leq 1$

Solve $\cos x = \dfrac{1}{2}$: Reference angle $\dfrac{\pi}{3}$. Cosine positive in QI and QIV.

$x = \dfrac{\pi}{3}, \quad x = 2\pi - \dfrac{\pi}{3} = \dfrac{5\pi}{3}$

$$\boxed{x = \frac{\pi}{3}, \frac{5\pi}{3}}$$

6. $2\cos^2\theta + \sin\theta = 1$ **on** $(0, 2\pi]$

Substitute $\cos^2\theta = 1 - \sin^2\theta$:

$2(1 - \sin^2\theta) + \sin\theta = 1$

$2 - 2\sin^2\theta + \sin\theta = 1$

$-2\sin^2\theta + \sin\theta + 1 = 0 \;\Rightarrow\; 2\sin^2\theta - \sin\theta - 1 = 0$

Let $u = \sin\theta$: $(2u+1)(u-1) = 0$

Case 1: $\sin\theta = 1 \;\Rightarrow\; \theta = \dfrac{\pi}{2}$

Case 2: $\sin\theta = -\dfrac{1}{2}$. Reference angle $\dfrac{\pi}{6}$. Sine negative in QIII and QIV.

$\theta = \pi + \dfrac{\pi}{6} = \dfrac{7\pi}{6}, \quad \theta = 2\pi - \dfrac{\pi}{6} = \dfrac{11\pi}{6}$

Interval $(0, 2\pi]$: all three solutions are in range ($\theta = 2\pi$ — check: $\sin 2\pi = 0 \neq -\frac{1}{2}$, so only 3 solutions).

$$\boxed{\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}}$$

7. $\tan^2 x = 3$ **on** $[0, 2\pi)$

Take square root: $\tan x = \pm\sqrt{3}$

Reference angle: $\dfrac{\pi}{3}$ (since $\tan\frac{\pi}{3} = \sqrt{3}$)

$\tan x = \sqrt{3}$: QI and QIII $\Rightarrow\ x = \dfrac{\pi}{3},\ \dfrac{\pi}{3} + \pi = \dfrac{4\pi}{3}$

$\tan x = -\sqrt{3}$: QII and QIV $\Rightarrow\ x = \pi - \dfrac{\pi}{3} = \dfrac{2\pi}{3},\ 2\pi - \dfrac{\pi}{3} = \dfrac{5\pi}{3}$

$$\boxed{x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}}$$

Part C — Multiple-Angle Equations

8. $\cos 2x = -1$ **on** $[0, 2\pi)$

Let $u = 2x$. Since $x \in [0, 2\pi)$, we have $u \in [0, 4\pi)$.

Solve $\cos u = -1$: $u = \pi,\ 3\pi$ (within $[0, 4\pi)$)

Convert back: $x = \dfrac{u}{2}$

$$x = \frac{\pi}{2}, \quad x = \frac{3\pi}{2}$$

$$\boxed{x = \frac{\pi}{2}, \frac{3\pi}{2}}$$

9. $\sqrt{3}\,\tan\left(\frac{x}{2}\right) - 1 = 0$ **on** $[0, 2\pi)$

Isolate: $\tan\left(\dfrac{x}{2}\right) = \dfrac{1}{\sqrt{3}}$

Let $u = \dfrac{x}{2}$. Since $x \in [0, 2\pi)$, we have $u \in [0, \pi)$.

Reference angle: $\dfrac{\pi}{6}$ (since $\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$)

Tangent positive in QI only (within $[0, \pi)$): $u = \dfrac{\pi}{6}$

Convert back: $x = 2u = 2 \cdot \dfrac{\pi}{6} = \dfrac{\pi}{3}$

$$\boxed{x = \frac{\pi}{3}}$$

10. $\sin 2x = \dfrac{1}{2}$ **on** $[0, 2\pi)$

Let $u = 2x$. Since $x \in [0, 2\pi)$, we have $u \in [0, 4\pi)$.

Reference angle: $\dfrac{\pi}{6}$. Sine positive in QI and QII.

In $[0, 2\pi)$: $u = \dfrac{\pi}{6}, \dfrac{5\pi}{6}$

In $[2\pi, 4\pi)$: $u = 2\pi + \dfrac{\pi}{6} = \dfrac{13\pi}{6}, \quad u = 2\pi + \dfrac{5\pi}{6} = \dfrac{17\pi}{6}$

Convert back $x = \dfrac{u}{2}$:

$$x = \frac{\pi}{12}, \quad x = \frac{5\pi}{12}, \quad x = \frac{13\pi}{12}, \quad x = \frac{17\pi}{12}$$

$$\boxed{x = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{13\pi}{12}, \frac{17\pi}{12}}$$